

Harpreet Singh

DATA ENGINEERING | PYTHON AND SQL | API AND SENSOR DATA PIPELINES

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Professional Summary

Research and embedded systems engineer building Python and SQL data pipelines for sensor streams, scholarly APIs, and documents. Experience with SQLite relational storage, schema design, joins, indexes, upserts, metadata normalization, and dataset validation. Develops acquisition, server-based data storage, batch preprocessing, and document-indexing workflows that produce reusable datasets for analysis and machine learning.

Technical Skills

Programming and databases: Python, SQL, SQLite, table design, primary keys, joins, indexes, parameterized queries, upserts, NumPy, Pandas.

Data pipelines and storage: API and stream ingestion, batch processing, server-based file storage, SQLite persistence, CSV/JSON/XDF, metadata normalization, deduplication.

Validation and automation: Schema checks, empty-dataset rejection, channel validation, automated unit tests, preprocessing manifests, scripted orchestration, diagnostic logging, Git.

Acquisition and integration: Lab Streaming Layer (LSL), LabRecorder, CGX EEG, UDP, HTTP, Raspberry Pi, Linux, SPI/GPIO, Zeek, Wireshark.

Selected Data Engineering Projects

CGX Dataset and Game | EEG acquisition and dataset pipeline

github.com/baazSingh13/CGX_dataset_game

- Built a Python ingestion workflow combining EEG streams and event markers in XDF; automated recording startup through service discovery, readiness checks, and background tasks.
- Saved acquired project data on a server; produced structured CSV datasets with timestamps, session/trial IDs, labels, and eight channels, plus JSON manifests linking source recordings and processing settings.
- Implemented required-column and empty-dataset checks, channel-count validation, and numeric conversion; preserved session, trial, and label boundaries when generating feature windows.
- Added automated tests for feature dimensions, finite preprocessing outputs, trial boundaries, and packet validation; connected batch preparation to training, held-session metrics, and model exports.

Sheeg | Recording orchestration and stream diagnostics

github.com/baazSingh13/Sheeg

- Scripted acquisition orchestration across CGX EEG, behavioural tasks, LSL markers, and LabRecorder; organized recording outputs by session with diagnostic logs.
- Documented a marker schema and added separate EEG/marker discovery diagnostics to investigate missing event streams and session-alignment failures.

ResearchTool | API ingestion and SQL data platform

github.com/baazSingh13/researchTool

- Built Python ingestion for scholarly metadata; normalized and deduplicated records into a persistent SQLite library with works, documents, chunks, and agent-run tables.
- Implemented parameterized SQL, primary-key upserts, work-key indexes, and document/chunk joins to support repeatable metadata loads and retrieval.
- Automated PDF extraction and chunk indexing; stored source paths, text/chunk counts, embeddings, and ingestion-run counts and errors for traceability.

Professional Experience

Research Assistant | University of Lethbridge | Apr 2025 - Present

- Develop research data workflows spanning EEG/fNIRS acquisition, behavioural event synchronization, labelled dataset creation, Python analysis, and model export.
- Build reusable preprocessing and integration scripts with schema checks, metadata manifests, automated tests, and live classification interfaces.
- Prepare temporal neural data for FPGA-oriented experiments, mapping 12 time bins by 196 features into fixed-point inputs with software reference checks.

Independent Embedded Systems Developer | Self-Employed | Jun 2024 - Mar 2025

- Designed a custom STM32F405RGT6 MicroPython development board and integrated sensor/actuator interfaces, firmware, and hardware debugging workflows.

Research Assistant | University of Regina | Jan 2021 - Apr 2024

- Prepared network-traffic features and Python machine-learning workflows for DoS/DDoS intrusion detection using Zeek, Wireshark, Bayesian Networks, and Markov Networks.
- Conducted preprocessing, feature engineering, model tuning, and experimental evaluation for uncertain-reasoning intrusion detection research published at SECRIPT 2024.
- Built additional medical-diagnosis and Punjabi NLP classification prototypes, including work with a corpus of more than 100,000 news articles.

Senior Software Engineer | Planetcast Media Services Ltd. | Dec 2018 - Nov 2020

- Led a 15-member team delivering embedded hardware and firmware for broadcast/media systems, including a network-controlled 10-channel video switcher and Ethernet temperature-acquisition platform.
- Developed Ethernet and HTTP/LWIP interfaces for remote monitoring, alarms, and device control, connecting physical measurements to networked applications.

Embedded Software Engineer / R&D Engineer | Spark Eighteen; Exicom | Feb 2016 - Dec 2018

- Developed STM32/AVR systems for LiFi, telecom power monitoring, interface cards, and automated test equipment; integrated CAN, SPI, I2C, and Ethernet interfaces.
- Performed board bring-up, firmware validation, and GUI-based test automation for industrial monitoring and control systems.

Education

PhD in Computational Behavioural Neuroscience | University of Lethbridge | In progress

M.Sc. Computer Science | University of Regina | 2024

Bachelor's equivalent in Electronics & Telecommunication Engineering | AMIETE/IETE | 2012

Research Publications

Singh, H., et al. "An Uncertain Reasoning-Based Intrusion Detection System for DoS/DDoS Detection." SECRIPT 2024, pp. 771-776.

Singh, H. A robust intrusion detection system utilizing uncertain reasoning techniques in artificial intelligence. M.Sc. thesis, University of Regina, 2024.